

Wazuh SOC Monitoring Enhancement Custom Dashboards, Email Alerting, and Telegram Notifications

1. Project Overview

Objective

Enhance Wazuh's monitoring and incident notification capabilities by implementing:

- Custom SOC dashboards
- Gmail alert notifications
- Telegram bot alert notifications
- Real-time validation using Sysmon-generated attack telemetry

The solution was tested against PowerShell-based attack activity and validated through Wazuh detections, dashboard visibility, email delivery, and Telegram notifications.

2. Lab Environment

Component	Technology
SIEM	Wazuh 4.x
Endpoint	Windows 11
Logging	Sysmon
Agent	Wazuh Agent
Email Alerting	Gmail SMTP
Mobile Alerting	Telegram Bot
Dashboard	Wazuh Dashboard

3. Dashboard Development

Objective

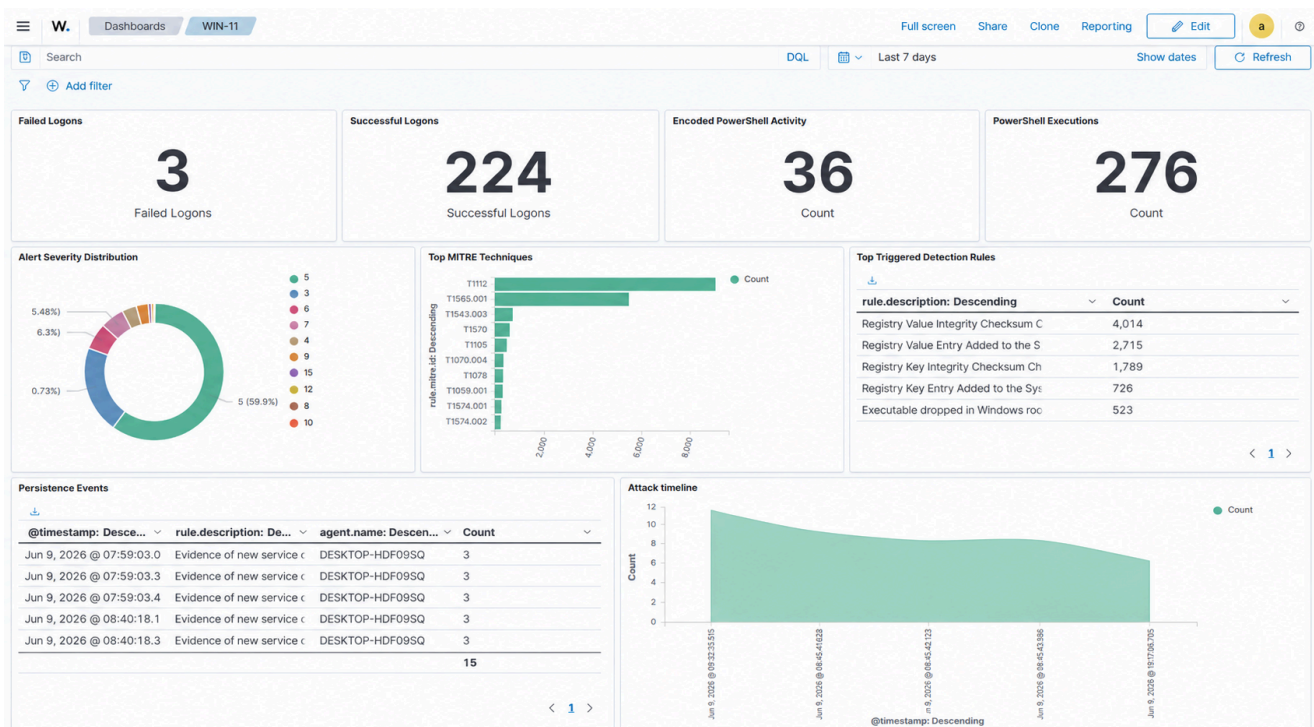
Create SOC-focused visualizations to improve threat visibility and investigation efficiency.

Dashboard Components

- Failed Logons (Event ID 4625)
- Successful Logons (Event ID 4624)
- Alert Severity Distribution
- MITRE ATT&CK Technique Distribution
- Encoded PowerShell Activity
- Persistence Monitoring
- Attack Timeline

Outcome

The dashboard provides centralized visibility into authentication activity, attack trends, MITRE ATT&CK mappings, and suspicious PowerShell execution.



4. Email Alerting Implementation

Objective

Provide near real-time notification of high-severity security alerts through email.

Configuration Files Modified

Wazuh Configuration

```
/var/ossec/etc/ossec.conf
```

```

GNU nano 0.3 /var/ossec/etc/ossec.conf
#-----
# Wazuh - Manager - Default configuration for anzn 2023
# More info at: https://documentation.wazuh.com
# Mailing list: https://groups.google.com/forum/#!forum/wazuh
#-----
<ossec_config>
  <global>
    <jsonout_output>yes</jsonout_output>
    <alerts_log>yes</alerts_log>
    <logall>no</logall>
    <logall_json>no</logall_json>
    <email_notification>yes</email_notification>
    <smtp_server>localhost</smtp_server>
    <email_from>edwindomini@mail.com</email_from>
    <email_to>edwindomini@mail.com</email_to>
    <email_maxperhour>60</email_maxperhour>
    <email_log_source>alerts.log</email_log_source>
    <agents_disconnection_time>15m</agents_disconnection_time>
    <agents_disconnection_alert_time>0</agents_disconnection_alert_time>
    <update_check>yes</update_check>
  </global>
  <alerts>
    <log_alert_level>3</log_alert_level>
    <email_alert_level>12</email_alert_level>
  </alerts>

```

Postfix Configuration

```

/etc/postfix/main.cf
/etc/postfix/sasl_passwd

```

```

relayhost = [smtp.gmail.com]:587
smtp_sasl_auth_enable = yes
smtp_sasl_password_maps = hash:/etc/postfix/sasl_passwd
smtp_sasl_security_options = noanonymous
smtp_tls_security_level = encrypt
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt

```

Email Delivery Architecture



Implementation Steps

Step 1

Installed Postfix and mail utilities.

Step 2

Configured Gmail SMTP relay.

Step 3

Generated Gmail App Password.

Step 4

Configured Postfix authentication.

Step 5

Configured Wazuh email notifications.

Step 6

Set alert threshold to Level 12 and above.

Step 7

Validated email delivery using Wazuh-generated alerts.

Result

High-severity Wazuh alerts were successfully delivered to Gmail.

Alerts included:

- Rule ID
- Severity Level
- Agent Name
- Event Details

- Sysmon Telemetry

W

Wazuh 3 days ago

to me ^

From Wazuh • edwindomir@gmail.com

To edwindomir@gmail.com

Date 10 Jun 2026, 09:51

Standard encryption (TLS).
[See security details](#)

Wazuh Notification.
2026 Jun 10 04:21:18

Received From: (DESKTOP-HDF09SQ) any->EventChannel
Rule: 92213 fired (level 15) -> "Executable file dropped in folder commonly used by malware"
User: DESKTOP-HDF09SQ\socadmin
Portion of the log(s):

```
{"win":{"system":{"providerName":"Microsoft-Windows-Sysmon","providerGuid":{"5770385f-c22a-43e0-bf4c-06f5698ffbd9"},"eventID":"11","version":"2","level":"4","task":"11","opcode":"0","keywords":"0x8000000000000000","systemTime":"2026-06-10T04:20:55.2000992Z","eventRecordID":"38549","processID":"3852","threadID":"3176","channel":"Microsoft-Windows-Sysmon/Operational","computer":"DESKTOP-
```

--END OF NOTIFICATION

Wazuh Notification.
2026 Jun 10 04:21:22

Received From: (DESKTOP-HDF09SQ) any->EventChannel
Rule: 92057 fired (level 12) -> "Powershell.exe spawned a powershell process which executed a base64 encoded command"
User: DESKTOP-HDF09SQ\socadmin
Portion of the log(s):

```
{"win":{"system":{"providerName":"Microsoft-Windows-Sysmon","providerGuid":{"5770385f-c22a-43e0-bf4c-06f5698ffbd9"},"eventID":"1","version":"5","level":"4","task":"1","opcode":"0","keywords":"0x8000000000000000","systemTime":"2026-06-10T04:20:57.8236558Z","eventRecordID":"38550","processID":"3852","threadID":"3176","channel":"Microsoft-Windows-Sysmon/Operational","computer":"DESKTOP-HDF09SQ","severityValue":"INFORMATION","message":"\\Process Create:\\nRuleName: -\\nUtcTime: 2026-06-10 04:20:57.796\\nProcessGuid: {842f9a06-e629-6a28-4902-000000004200}\\nProcessId: 9344\\nImage: C:\\Windows\\System32\\Windows PowerShell\\v1.0\\powershell.exe\\nFileVersion: 10.0.26100.5074
```

5. Telegram Alerting Implementation

Objective

Provide mobile-based alert delivery for critical security events.

Configuration Files Modified

Wazuh Integration

```
/var/ossec/etc/ossec.conf
```

Custom Integration Script

```
/var/ossec/integrations/telegram
```

Telegram Alert Architecture



Implementation Process

Step 1

Created Telegram Bot using BotFather.

Step 2

Obtained:

- Bot Token
- Chat ID

Step 3

Developed custom Python integration.

The script:

- Receives Wazuh JSON alerts
- Extracts agent information
- Extracts rule information
- Extracts severity level

- Sends formatted alert to Telegram

```
GNU nano 8.3 /var/ossec/integrations/telegram
#!/usr/bin/env python3

import json
import requests
import sys

TOKEN = " "
CHAT_ID = " "

alert_file = sys.argv[1]

with open(alert_file) as f:
    alert = json.load(f)

rule_id = alert['rule']['id']
level = alert['rule']['level']
description = alert['rule']['description']
agent = alert['agent']['name']

message = (
    f"!!! WAZUH ALERT !!! \n\n"
    f"Agent: {agent}\n"
    f"Rule: {rule_id}\n"
    f"Level: {level}\n"
    f"Description: {description}"
)

url = f"https://api.telegram.org/bot({TOKEN})/sendMessage"

response = requests.post(
    url,
    data={
        "chat_id": CHAT_ID,
        "text": message
    }
)

print(response.status_code)
print(response.text)
```

Step 4

Configured Wazuh Integration Block

```
<integration>
  <name>telegram</name>
  <level>12</level>
  <alert_format>json</alert_format>
</integration>
```

Step 5

Validated alert delivery.

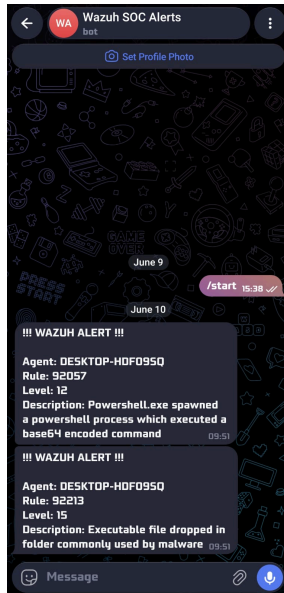
Result

Telegram notifications were successfully delivered for Level 12+ security alerts.

Delivered information:

- Agent Name
- Rule ID
- Severity Level

- Alert Description



6. Attack Validation

Attack Executed

```
powershell.exe -nop -w hidden -ep bypass -EncodedCommand  
VwByAGkAdABlAC0ATwB1AHQAcAB1AHQAIAAiAEgAZQBsAGwAbwAiAA==
```

Purpose:

Generate realistic PowerShell telemetry for validation of detection and alerting workflows.

7. Detection Results

Detection 1

Rule ID 92057

Description:

```
Powershell.exe spawned a powershell process which executed a base64 encoded  
command
```

Severity:

12

MITRE ATT&CK:

T1059.001 – PowerShell

Detection Source:

Sysmon Event ID 1

Detection 2

Rule ID 92213

Description:

Executable file dropped in folder commonly used by malware

Severity:

15

MITRE ATT&CK:

T1105 – Ingress Tool Transfer

Detection Source:

Sysmon Event ID 11

```
> Jun 10, 2026 @ 09:51:22.316 {input.type: log agent.ip: 192.168.56.106 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.image: C:\\WINDOWS\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.processGuid: {84
2f9a06-e629-6a28-4902-000000004200} data.win.eventdata.processId: 9344 data.win.eventdata.utcTime: 2026-06-10 04:20:57.913
data.win.eventdata.targetFileName: C:\\Users\\socadmin\\AppData\\Local\\Temp\\__PSScriptPolicyTest_hboekza.hnx.ps1
data.win.eventdata.creationUtcTime: 2026-06-10 04:20:57.913 data.win.eventdata.user: DESKTOP-HDF09SQ\\socadmin

> Jun 10, 2026 @ 09:51:22.314 {input.type: log agent.ip: 192.168.56.106 agent.name: DESKTOP-HDF09SQ agent.id: 001 manager.name: wazuh-server
data.win.eventdata.originalFileName: PowerShell.EXE data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0
\\powershell.exe data.win.eventdata.product: Microsoft® Windows® Operating System data.win.eventdata.parentProcessGuid: {842
f9a06-e626-6a28-4702-000000004200} data.win.eventdata.description: Windows PowerShell data.win.eventdata.logonGuid: {842f9a0
6-daf3-6a28-d661-0a0000000000} data.win.eventdata.parentCommandLine: \\\"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powers
```

8. Results

The implementation successfully achieved:

- Centralized security monitoring
- Custom SOC dashboard visibility
- Email-based alert delivery
- Telegram-based mobile alert delivery

- Validation against Sysmon attack telemetry
 - MITRE ATT&CK mapped detections
- Both notification channels successfully delivered real-time alerts for Rules 92057 and 92213.
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9. Conclusion

This project enhanced a Wazuh-based SOC environment by implementing custom monitoring dashboards and multiple alert delivery mechanisms. The solution was validated using encoded PowerShell execution activity captured by Sysmon and correlated by Wazuh. Successful delivery of alerts through Gmail and Telegram demonstrates a practical incident notification workflow suitable for SOC monitoring and threat detection environments.
